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PROCESSES FOR MAKING VINYL ACETATE FROM ETHYLENE GLYCOL DIACETATE

Abstract

The various embodiments disclosed herein relate to a process for producing vinyl acetate monomer (VAM) from ethylene glycol diacetate (GDA) by thermal cracking in the presence of a substantially non-reactive liquid. The liquid phase enhances heat transfer, improves selectivity, and reduces fouling. The process operates in a continuous or discontinuous phase, optimizing efficiency and enabling higher yields of VAM with improved reactor performance.

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Background/Summary

CROSS-REFERENCES & RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/561,119 filed Mar. 4, 2024 and entitled "PROCESSES FOR MAKING VINYL ACETATE FROM ETHYLENE GLYCOL DIACETATE," which is hereby incorporated by reference in its entirety under 35 U.S.C. § 119 (e).

BACKGROUND

[0002] Vinyl acetate monomer (VAM) is an important commodity chemical. Numerous processes exist for making VAM. An interest exists in using sustainable feedstocks rather than fossil-based feedstocks to make commodity chemicals, including VAM. Commercial acceptance of commodity chemicals made using sustainable feedstocks will, in part, depend on the cost of their production.

[0003] Vinyl acetate is typically made by the catalytic reaction of ethylene, acetic acid and oxygen. Some industrial production uses the hydroesterification of acetic acid and acetylene. A proposed route is the thermal or catalytic cracking of 1,1-diacetoxyethane. See, for instance, U.S. Pat. No. 2,425,389 and European Patent Application Publication No. 0 048 173 A1. The high costs of producing 1,1-diacetoxyethane have not led to this proposed route achieving commercial viability for making VAM.

[0004] Vinyl acetate monomer can also be made from ethylene glycol diacetate. U.S. Pat. Nos. 2,251,983; 3,689,535; 3,787,485 and 3,804,887 disclose the thermal cracking of ethylene glycol diacetate to vinyl acetate. The thermal cracking of ethylene glycol diacetate has also not seen commercial viability.

[0005] Nevertheless, providing an economically attractive route for making vinyl acetate monomer from ethylene glycol diacetate is desired as the ethylene glycol diacetate can be obtained from renewable as well as fossil-based feedstocks. Making vinyl acetate monomer from sustainable resources can proceed through several routes. For instance, ethylene glycol diacetate can be made by the esterification of monoethylene glycol. The conventional process for making monoethylene glycol is by cracking fossil-based feedstock to produce ethylene, partially oxidizing the ethylene-to-ethylene oxide and then hydrolyzing ethylene oxide to monoethylene glycol. The conventional process could use ethylene that is derived from sustainable feedstocks, e.g., ethylene made by the dehydration of ethanol derived from the fermentation of carbohydrate. Other processes exist for making monoethylene glycol from sustainable feedstocks. For instance, sustainable feedstock can be converted to syngas and monoethylene glycol can be made by the glycol oxalate process such as disclosed in U.S. Pat. No. 4,453,026. Also, recent efforts have been devoted to converting carbohydrates to monoethylene glycol by retro aldol conversion of the carbohydrate to, among others, glycol aldehyde, and hydrogenation to monoethylene glycol. See, for instance, U.S. Pat. No. 9,783,472.

[0006] The commercial viability of processes for making vinyl acetate monomer from ethylene glycol diacetate, whether from fossil sources or sustainable resources, will depend upon, among other things, the efficiencies of converting ethylene glycol diacetate to vinyl acetate monomer.

BRIEF SUMMARY

[0007] This invention pertains to processes for making vinyl acetate monomer from ethylene glycol diacetate wherein ethylene glycol diacetate is contacted with a continuous or discontinuous liquid phase of a substantially non-reactive component, which component is at a temperature sufficient to convert ethylene glycol diacetate to vinyl acetate monomer and acetic acid ("cracking temperature"). By this invention it is believed that a number of advantages can be achieved. Most notably, the contact of ethylene glycol diacetate and vinyl acetate monomer at cracking temperatures with metal surfaces that can promote undesired reactions or become fouled, can be maintained at a low incidence. Thus, enhanced selectivities to vinyl acetate monomer and reduced reactor maintenance costs to address fouling, can be achieved. Moreover, as the conversion of ethylene glycol diacetate to vinyl acetate monomer is endothermic, the continuous or discontinuous liquid phase provides heat proximate to the sites of the cracking thereby enhancing conversion rates. The ability to provide heat proximate to the sites of the cracking liquid phase enables a relatively uniform temperature profile in the reaction zone as compared to conventional thermal cracking processes where the heat is only provided at the walls of the reaction zone. Thus, the operator has better control of the process for optimization of production of vinyl acetate monomer. In some instances, enhanced net production of vinyl acetate monomer per unit of ethylene glycol acetate fed to the reaction zone can be achieved as compared to thermal cracking where the heat is provided by the walls of the reactor all other conditions being substantially the same.

[0008] In the broad aspects, the processes of this invention comprise: continuously supplying ethylene glycol diacetate to a reaction zone; contacting in the reaction zone the ethylene glycol diacetate with a substantially non-reactive liquid, said liquid being (i) at a temperature higher than the boiling point at the pressure in the reaction zone of the ethylene glycol diacetate supplied such that a mixed liquid-vapor phase exists in the reaction zone, and (ii) at a temperature sufficient to convert at least a portion of the ethylene glycol diacetate to vinyl acetate monomer and acetic acid and provide a product gas comprising vinyl acetate monomer, acetic acid and unreacted ethylene glycol diacetate; and continuously withdrawing said product gas from the reaction zone.

[0009] The ethylene glycol diacetate can be supplied to the reaction zone as a liquid or a vapor. Where supplied as a liquid, the ethylene glycol diacetate vaporizes. Thus, in either event a vapor phase and a liquid phase will exist in the reaction zone. In some instances, the ethylene glycol diacetate is preheated to a temperature between about 250° C. and 450° C., say, 350° C. to 425° C., which is below a temperature where significant cracking occurs, and the ethylene glycol diacetate is already in the vapor phase prior to contact with the liquid. Thus, the liquid need not provide the latent heat of vaporization and only needs to provide the sensible heat required to bring the ethylene glycol diacetate to reaction temperature.

[0010] Often the liquid is at a temperature within the range of about 400° C. to 570° C., say, 450° C. to 550° C. The amount of liquid used, in addition to providing any latent heat for vaporization or sensible heat for raising the temperature of the ethylene glycol diacetate, is sufficient to attenuate the loss of heat in the region proximate to a molecule of ethylene glycol diacetate undergoing cracking to generate vinyl acetate and acetic acid. The liquid phase can be the continuous phase or the discontinuous phase in the reaction zone. Where the liquid phase is the discontinuous phase, it is preferred that the liquid phase be in a relatively uniform distribution in at least a portion of the cross-section of the reaction zone perpendicular to the direction of flow of the vaporous phase passing through the reactor. Operations using the liquid as the continuous phase are typically preferred. Often the design of the reaction zone is such that the ethylene glycol diacetate is passed to a lower portion of the liquid phase as small bubbles, say, less than 2 centimeters in diameter, preferably less than 5 millimeters, and sometimes less than 20 microns in diameter.

[0011] Most frequently only a portion of the ethylene glycol diacetate fed to the reaction zone is converted, say, less than 75 percent, and often between about 15 and 50 percent, is converted. In such cases, the product gas withdrawn from the reaction zone would contain unreacted ethylene glycol diacetate. The product gas can be cooled to a temperature to condense at least a portion of the unreacted ethylene glycol diacetate but maintain the vinyl acetate monomer in the vaporous phase, the condensate separated from the vaporous phase and at least a portion of the condensate is recycled to reaction zone.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIGS. **1** and **2** are schematic depictions of an apparatus that can be used in the processes of the claimed processes.

[0013] FIG. **3** is a schematic depiction of an injector that can be used to contact the non-reactive liquid with ethylene glycol diacetate which is either in the liquid phase or vaporous phase.

DETAILED DESCRIPTION

[0014] All patents, published patent applications, and articles referenced herein are hereby incorporated by reference in their entirety.

Definitions

[0015] As used herein, the following terms have the meanings set forth below unless otherwise stated or clear from the context of their use.

[0016] Where ranges are used herein, the end points only of the ranges are stated so as to avoid having to set out at length and describe each and every value included in the range. Any appropriate intermediate value and range between the recited endpoints can be selected. By way of example, if a range of between 0.1 and 1.0 is recited, all intermediate values (e.g., 0.2, 0.3, 0.63, 0.815 and so forth) are included as are all intermediate ranges (e.g., 0.2-0.5, 0.54-0.913, and so forth).

[0017] The use of the terms “a” and “an” is intended to include one or more of the elements described.

[0018] Admixing or admixed means the formation of a physical combination of two or more elements which may have a uniform or non-uniform composition throughout and includes, but is not limited to, solid mixtures, solutions and suspensions.

[0019] Ethylene glycol acetate is also known as 1,2-diacetoxyethane and ethylene glycol diacetate is abbreviated herein as GDA.

[0020] The half acetate ester of ethylene glycol is also known as 2-hydroxyethyl acetate and is abbreviated herein as HEA.

[0021] Substantially non-reactive means that under the reaction conditions including time, temperature and presence of catalyst and other adjuvants, less than 1 percent of the moiety would be reacted.

[0022] Sustainable resources are plants and animals, including, but not limited to, waste products from plants and animals, and sustainable feedstocks are derived from sustainable resources. Sustainable resources also include carbon dioxide used as a feedstock in processes to make monoethylene glycol, whether that carbon dioxide is captured from direct air capture or from emissions from facilities that emit carbon dioxide, including, but not limited to, incineration, power generation, fermentations, and chemical and other industrial processes.

[0023] According to various implementations, ethylene glycol diacetate in the vaporous phase can be converted to vinyl acetate monomer and acetic acid in the presence of a substantially non-reactive liquid. In these implementations, the liquid has a boiling point under the pressure in the reaction zone equal to, or preferably higher; in one example, at least about 10° C. higher than the bulk temperature in the reaction zone. Examples of liquids include higher molecular weight hydrocarbons which may be aliphatic or aromatic, with 30 or more carbons. Useful liquids include paraffins of 36 or more carbons, especially normal paraffins of 36 to 50 carbons. Liquids in which acetic acid has limited solubility, such as higher molecular weight paraffins, are particularly advantageous.

[0024] In some implementations, the amount of liquid used is sufficient to attenuate the loss of heat proximate to a site where ethylene glycol diacetate is being cracked to vinyl acetate and acetic acid. A number of factors may be taken into consideration when determining the amount of liquid to be used. In implementations where the liquid is the continuous phase, the primary variables are the temperature of the liquid and the desired degree of conversion of ethylene glycol diacetate. In one exemplary implementation, at lower liquid temperatures, the rate of conversion may be slower, and thus for increased conversion, more liquid would be used. Also, the geometric configuration of and flow path of bubbles of ethylene glycol diacetate of the reaction zone may be a factor. In some implementations where the liquid is a discontinuous phase, additional factors may need to be taken into account such as the extent that the walls of the reaction zone provide heating or cooling and the duration of contact between the liquid and vaporous phases. As would be understood, the rate of supply of the ethylene glycol diacetate to the reaction zone may be such that at least 2, optionally at least 5, grams of liquid phase are present in the reaction zone per gram of ethylene glycol diacetate fed per second to the reaction zone. In some implementations, especially where the liquid phase is the continuous phase, at least 50, such as, 100 to 5000, grams of liquid phase are present in the reaction zone per gram of ethylene glycol diacetate fed per second to the reaction zone.

[0025] The ethylene glycol diacetate fed to the reaction zone can be about all ethylene glycol diacetate or be in combination with one or more components, for instance, propylene glycol diacetate and/or 1,2-diacetoxybutane. Other components can include, but are not limited to, diluents such as inert gases and hydrocarbons such as methane and ethane. Cracking can provide acetic acid and vinyl acetate and, if propylene glycol diacetate is in the feedstock, allyl acetate, and if 1,2-diacetoxybutane is in the feedstock, acetoxybutene such as one or more of 1-acetoxy-2-butene, 2-acetoxy-1-butene, and 2-acetoxy-2-butene. A vaporous phase comprising these components may be withdrawn from the reaction zone and subjected to distillation to provide a vinyl acetate rich stream. Preferably the vinyl acetate rich stream contains less than 0.1, most preferably less than 0.01, mass percent allyl acetate and less than 50 parts per million by mass of any compound containing four or more carbons bonded in series.

[0026] The temperatures in the reaction zone may be in the range of about 350° C. to 750° C., say, between about 450° C. to about 550° C., are often employed, with higher temperatures being used in combination with higher space velocities than can be used at lower temperatures. Residence times at higher temperatures, e.g., above about 450° C., may be limited to attenuate adverse reactions such as the generation of ethylene from the vinyl acetate. In some implementations, the time that the vaporous phase is at a temperature of above about 350° C. is less than about 200 seconds, and frequently less than about 50 seconds, or even less than about 10 seconds. In some implementations with temperatures above about 450° C. are used, the duration that the vaporous phase is at or above about 450° C. is often less than about 10 seconds, and sometimes less than about 5 seconds. Dilution of the ethylene glycol diacetate in the vaporous phase with an inert, e.g., nitrogen, methane, and ethane, where the cracking is in the vaporous phase, can assist in achieving short residence times. The reaction zone pressure can be in the range of below atmospheric to 5000 kPa or more, and often in the range of about 50 to 2500 kPa absolute. Higher pressures serve to increase the boiling point of the non-reactive liquid thereby enabling the use of liquids that would otherwise boil and serve to reduce the fraction of liquid in the vapor phase.

[0027] It would be appreciated that with very short residence times at temperature, only a portion of the ethylene glycol diacetate would be converted. In some implementations, the conversion of ethylene glycol diacetate is between about 10 percent and about 75 percent, say between about 15 percent and about 50 percent. Generally, at the lower conversions, the selectivity to vinyl acetate is enhanced, all else being substantially constant. Preferably, the thermal cracking temperature, space velocity and time at temperatures above 350° C., provide a selectivity of ethylene glycol diacetate converted to vinyl acetate of at least about 60, and sometimes at least about 70, percent. The ethylene glycol diacetate not converted can be recycled, after removal of vinyl acetate, to the reaction zone.

[0028] Any suitable reactor or series of reactors can be employed in conducting the cracking, and the process can be batch, semi-continuous or, preferably, continuous. A reactor can be one or more vessels in series or in parallel and a vessel can contain one or more zones. A reactor can be of any suitable design for continuous operation including, but not limited to, tanks and pipe or tubular reactor and can have, if desired, fluid mixing capabilities. Types of reactors include, but are not limited to, tubular reactors, stirred tank reactors, trickle-bed reactors, falling film reactors, bubble column, sieve trayed column, valve trayed column, packed and loop reactors. In some instances, the cracking may be conducted in whole or part in a reactive distillation apparatus used to make the ethylene glycol diacetate. In some instances, the cracking may be conducted in a separate reactive distillation apparatus from that used to make the ethylene glycol diacetate. It should be understood that the cracking may be conducted in one or more zones, each being at the same or being at different cracking conditions, including, but not limited to, the same or different residence times, pressures, and temperatures.

[0029] The heavies from the cracking contain unreacted ethylene glycol diacetate and other high boiling esters, if present, such as propylene glycol diacetate and 1,2-acetoxybutane. Often the distillation is designed to obtain an off-take of ethylene glycol diacetate and other diacetates, if present, for recycle to the reaction zone. The diacetate fraction can be withdrawn from the distillation as a liquid phase or as a vapor phase.

DRAWINGS

[0030] Reference is made to the drawings which are provided to facilitate the understanding of the instantly disclosed process but are not intended to be in limitation of the disclosed process. The drawings omit ancillary unit operations and omit minor equipment such as pumps, heat exchangers, valves, instruments and other devices and unit operations the placement of which, and operation thereof, are well known to those practiced in chemical engineering.

[0031] FIG. **1** is a schematic depiction of an apparatus generally designated as **100** suitable for practicing the described processes where the non-reactive liquid is the continuous phase in the reaction zone. The apparatus comprises reactor **102** which for the purposes of discussion can be an unstirred tank containing a liquid phase **104**, or just liquid **104**, at the bottom. For purposes of this discussion the liquid is tetracontane having a melting point of about 80° C. to 84° C. and a normal boiling point of about 524° C. For purposes of discussion, the reaction zone is maintained with a head pressure of about 200 kPa absolute. The reactor **102** can be jacketed with heating coils such as high-pressure steam or electrical resistance which can be used to at least melt the tetracontane. As shown, line **106** continuously withdraws tetracontane, passes it through furnace **108** and returns the heated tetracontane to the reactor **102**. For purposes of discussion, the liquid in the reactor **102** is maintained at substantially 500° C., the tetracontane from furnace **108** is at a temperature of about 525° C. and the rate of tetracontane flow in line **106** is sufficient to provide liquid **104** at a bulk temperature of 500° C. The tetracontane in line **106** is preferably introduced at multiple points in the reactor **102** to assist in achieving a more uniform liquid temperature.

[0032] A feed of ethylene glycol diacetate is passed via line **110** to indirect heat exchanger **112** where it is vaporized and heated to a temperature below that where appreciable thermal cracking of the ethylene glycol diacetate would occur, typically in the range of about 300° C. to 400° C. As shown, the heat exchange is with tetracontane withdrawn from reactor **102** via line **114** and passed to heat exchanger **112**. As shown, line **114** withdraws liquid from an upper region of the reactor to minimize the concentration of vinyl acetate, acetic acid and unreacted ethylene glycol diacetate in the liquid.

[0033] The vaporized and heated ethylene glycol diacetate is passed from heat exchanger **112** via line **116** to venturi mixer **118**. The cooler tetracontane from heat exchanger **112** is passed via line **120** to venturi mixer **118** where it is used as a motive fluid. In venturi mixer **118**, the vaporous phase containing ethylene glycol diacetate is drawn into the motive fluid as a finely divided dispersion and this mixed phase fluid is passed to nozzle **122** and introduced into a lower portion of liquid **104**. Venturi mixer **118** is thus part of a vapor injection system to provide finely dispersed bubbles. With small bubbles, the bubbles tend to follow the flow of the liquid rather than taking a direct route upwardly. Often, the bubbles are less than 20 microns in diameter. Many types of vapor injectors are known and used industrially. One preferred type of vapor injector is described in U.S. Pat. No. 4,162,970. A plurality of vapor injectors can be used to distribute ethylene glycol diacetate over the cross-section of liquid **104**. The injection of the motive fluid and the vaporous ethylene glycol diacetate results in agitation of liquid **104**. Hence mechanical stirring of liquid **104** typically is not used. Where the bubbles are sufficiently fine that they tend to follow the flow of the liquid, agitation can be beneficial to increase residence time of the ethylene glycol diacetate in the reaction zone.

[0034] The depth of liquid **104** together with the rate of upwardly passing bubbles define the average residence time of the vaporous phase in the reaction zone. Thus, the operator can use the depth of liquid **104** to control the duration of the cracking reaction. The vapor in the head space above the surface of liquid **104** will contain vinyl acetate, acetic acid, unreacted ethylene glycol diacetate and reaction side products. This vaporous phase is withdrawn via line **124** and chilled in heat exchanger **126**, e.g., with water supplied via line **128**, to a temperature below that which thermally cracks ethylene glycol diacetate, e.g., below 350° C., but preferably sufficiently high to maintain the moieties in the stream in the vapor phase. The chilled vaporous phase is passed via line **130** to rectification column **132** where an overhead of vinyl acetate monomer, acetic acid and other lights is withdrawn via line **134** for further purification of the vinyl acetate monomer. A bottoms fraction containing ethylene glycol diacetate is withdrawn via line **136** and is passed to heat exchanger **112** for recycle.

[0035] A purge stream **138** is taken from line **114** to avoid the build up of undesired high boiling contaminants, such as polyvinyl acetate, in liquid **104**. A purge stream **140** is taken from line **136** to avoid the build-up of contaminants in the ethylene glycol diacetate fraction.

[0036] With respect to FIG. **2**, a schematic depiction of apparatus **200** in which the non-reactive liquid is in the discontinuous phase is presented. Apparatus **200** comprises reactor **201** having contact zone **202** and, at the top, disentrainment zone **203**. An ethylene glycol diacetate-containing feed is supplied to apparatus **200** via line **204**. The feed is heated and vaporized in heat exchanger **206**, which for purposes of discussion is 400° C. The vaporized feed is passed from heat exchanger **206** via line **208** to distribution nozzle **210** positioned in a lower portion of contact zone **202**. The vaporous phase passes upwardly in the contact zone and is contacted with droplets of the non-reactive liquid which passes counter currently. The liquid at the bottom of contact zone **202** is withdrawn from reactor **201** via line **212** and is used for heating the incoming feed in heat exchanger **206**. The liquid is then passed via line **214** to furnace **216** where it is heated to, say, 525° C. The heated liquid then passes via line **218** to distributor **220** positioned in an upper portion of contact zone **202**. Distributor **220** may be of any suitable design. Typically, the droplet size is less than 10, preferably less than 2, millimeters in diameter. For purposes of discussion, the droplet size is about 1 millimeter in diameter. The amount of liquid used is sufficient to heat the incoming feed to about 500° C. Distributor **220** provides a relatively uniform distribution of liquid droplets over the cross section of contact zone **202**. The rate of supply of the liquid is such that at any given time, the volume of contact zone **202** that contains liquid is about 5 percent. The distribution of the droplets taken together with their rate of descent in contact zone **202** provides liquid proximate to ethylene glycol diacetate molecules undergoing cracking and thereby attenuates heat loss proximate to the endothermic reaction. The duration of the feed in contact zone **202** is sufficient to provide the conversion of a portion of the ethylene glycol diacetate, say, between about 15 and 35 percent.

[0037] The vaporous phase exits contact zone **202** and enters disentrainment zone **203**. Disentrainment zone **203** has a larger cross-sectional area and the drop in vapor velocity allows entrained liquid to fall out. At least a portion of the liquid is directed by the disentrainment zone to the sides of contact zone **202**, thereby limiting contact of the ethylene glycol diacetate and reaction products with the walls. While the walls of contact zone **202** may be heated, it usually is not necessary as the heat is provided by the non-reactive liquid.

[0038] A vaporous phase is withdrawn from an upper portion of disentrainment zone **203** via line **232** and passed to heat exchanger **224** where it is chilled, e.g., with water supplied via line **226**, to a temperature below that which thermally cracks ethylene glycol diacetate, e.g., below 350° C., but preferably sufficiently high to maintain the moieties in the stream in the vaporous phase. The chilled vaporous phase is passed via line **228** to rectification column **230** where an overhead of vinyl acetate monomer, acetic acid and other lights is withdrawn via line **232** for further purification of the vinyl acetate monomer. A bottoms fraction containing ethylene glycol diacetate is withdrawn via line **234** and is passed to heat exchanger **206** for recycle.

[0039] A purge stream **236** is taken from line **214** to avoid the buildup of undesired high boiling contaminants, such as polyvinyl acetate, in the liquid. A purge stream **238** is taken from line **234** to avoid the build-up of contaminants in the ethylene glycol diacetate fraction.

[0040] With respect to FIG. **3** an injector **300** is depicted for the admixing of ethylene glycol diacetate feed and non-reactive liquid. The injector **300** can serve as all or a part of the reaction zone. The ethylene glycol diacetate feed is passed through line **302** to a throat region **304** of the injector. Hot liquid, e.g., at 550° C., is passed via line **306** into injector **300** and passes through an orifice plate **308** into throat region **304**. As shown, when the hot liquid passes through orifice plate **308**, a low-pressure region **312** is formed which assists in the admixing of the ethylene glycol feed and the hot liquid. If the feed is a liquid, it is vaporized and forms a two-phase stream upon vaporization of the feed in which the hot liquid is the continuous phase. If the feed is vaporous, the low pressure facilitates the inclusion of the feed as fine bubbles in the hot liquid.

[0041] For purposes of discussion, the two-phase mixture is at a temperature of about 500° C. which enables cracking of the ethylene glycol diacetate to vinyl acetate and acetic acid. Throat region **304** may be sufficient in length to enable a desired degree of conversion, i.e., the throat region serves in essence as a tubular reactor. In this case, the throat region **304** directs the mixture to a disentrainment zone **309** within the reactor vessel **310** for liquid and vapor phase separation. As discussed before, the liquid can be heated and recycled. The vaporous phase can be processed to recover vinyl acetate and unreacted ethylene glycol diacetate, which can be recycled. Alternatively, throat region **304** can direct the mixture to a reactor vessel **310** for continuing the reaction of ethylene glycol diacetate to vinyl acetate and acetic acid. The vinyl acetate product can be obtained from the vaporous effluent from reactor vessel **310** in the manner described above in respect of FIGS. **1** and **2**.

[0042] Although the disclosure has been described with reference to preferred implementations, persons skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the disclosed apparatus, systems and methods.

Claims

1. A process for converting ethylene glycol diacetate to vinyl acetate comprising: continuously supplying ethylene glycol diacetate to a reaction zone, said ethylene glycol diacetate; contacting in the reaction zone the ethylene glycol diacetate with a substantially non-reactive liquid, said liquid being (i) at a temperature higher than the boiling point at the pressure in the reaction zone of the ethylene glycol diacetate supplied such that a mixed liquid-vapor phase exists in the reaction zone, and (ii) at a temperature sufficient to convert at least a portion of the ethylene glycol diacetate to vinyl acetate monomer and acetic acid and provide a product gas comprising vinyl acetate monomer, acetic acid and unreacted ethylene glycol diacetate; and continuously withdrawing said product gas from the reaction zone.

2. The process of claim 1 wherein the non-reactive liquid is provided as a continuous phase in the reaction zone.

3. The process of claim 1 wherein the non-reactive liquid is provided as a discontinuous phase in the reaction zone.

4. The process of claim 1 wherein the non-reactive liquid is at a temperature between about 450° C. and 550° C.

5. The process of claim 1 wherein the ethylene glycol diacetate supplied is vaporous.
6. The process of claim 1 wherein between about 15 and 50 percent of the ethylene glycol diacetate is converted in the reaction zone.
7. The process of claim 1 wherein the non-reactive liquid has a boiling point under the pressure in the reaction zone at least about 10° C. higher, than the temperature in the reaction zone.
8. The process of claim 1 wherein the non-reactive liquid is an aromatic or aliphatic hydrocarbon of at least 30 carbon atoms.
9. The process of claim 1 wherein the time that the vaporous phase is at a temperature of above 350° C. is less than about 50 seconds.
10. The process of claim 1 wherein non-reactive liquid is at a temperature between about 450° C. and 550° C. and the time that the vaporous phase is at a temperature of above 350° C. is less than about 10 seconds.
11. The process of claim 1 wherein the non-reactive liquid is provided as a continuous phase in the reaction zone and the vaporous phase comprises bubbles less than 5 millimeters in diameter.
12. The process of claim 11 wherein the vaporous phase bubbles are less than 1 millimeter in diameter.
13. The process of claim 1 wherein the ethylene glycol diacetate is supplied as a vapor and at a temperature between about 250° C. and 450° C.
14. The process of claim 1 wherein the ethylene glycol diacetate is supplied as a liquid and is vaporized by contact with the non-reactive liquid.